

Worked Example — The 6-Vendor Anomaly (Vendors, the 18th transaction type)

Every equation NGAME runs, by example, from raw snapshot to the dashboard's HIGH warning

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Worked Example — The 6-Vendor Anomaly (φ_{18})

A bookkeeper added **6 new vendors in a single day**. This note traces every equation NGAME executes on the real numbers from `NGAME_Fraud_Analysis_readable_truly_clean.json` and the Vendors row of `NGAME_Training_Matrix.xlsx` — each value plugged in — to show exactly how the dashboard arrives at **8.9% · $z \approx -7.28$ · HIGH**, ranked first of all 18 transaction types.

What the -7.28 means (the short answer)

The -7.28 is the **z -score**: today's Vendors CPI sits **7.28 standard deviations below** the mean stability NGAME learned for Vendors during training. Anything past 3σ is astronomically unlikely under normal operation — hence HIGH. The 8.9% shown beside it is a softer, secondary readout (how far below normal as a plain percentage); the -7.28 is what the alarm logic keys on.

The inputs (pulled straight from the files)

Quantity	Symbol	Value	Source
Today's Vendors CPI	CPI_{18}	0.90625	<code>today_cpi_array</code> → Vendors (JSON)
Trained mean CPI	μ_{18}	0.994794174882256	Vendors row, col B (matrix)
Trained std deviation	σ_{18}	0.01216178288304404	Vendors row, col C (matrix)
New vendors added today	c_{18}	6	stated scenario
Vendors removed today	b_{18}	0	stated scenario

Step 1 — Daily set-up: recover the contingency counts

The dashboard stores the resulting CPI, not the raw counts. But with 6 added and 0 removed, the counts are forced by the CPI value. Solve $\text{CPI} = a/(a + b + c)$ for a :

$$0.90625 = \frac{a}{a + 0 + 6}$$

$$0.90625(a + 6) = a \Rightarrow 0.90625a + 5.4375 = a$$

$$5.4375 = a - 0.90625a = 0.09375a \Rightarrow a = \frac{5.4375}{0.09375} = 58$$

$$a_{18} = 58 \text{ retained, } b_{18} = 0 \text{ removed, } c_{18} = 6 \text{ added, } n_{18} = 58 + 0 + 6 = 64$$

Step 2 — Daily per-type measures: CPI and its companions

$$\text{CPI}_{18} = \frac{a}{n} = \frac{58}{64} = 0.90625$$

$$\text{AdditionRate}_{18} = \frac{c}{n} = \frac{6}{64} = 0.09375 \quad \text{DeletionRate}_{18} = \frac{b}{n} = \frac{0}{64} = 0$$

$$0.90625 + 0.09375 + 0 = 1.0 \quad \checkmark$$

Already the signal is visible: nearly **one in ten** vendors in today's ledger did not exist yesterday (AdditionRate = 9.375%).

Step 3 — Percent deviation (the 8.9% on the dashboard)

A plain “how far below normal” readout — the absolute gap from the mean, expressed as a percent of the mean.

$$\text{absolute_difference} = |\text{CPI}_{18} - \mu_{18}| = |0.90625 - 0.994794174882256| = 0.088544174882256$$

$$\text{percent_deviation} = \frac{\text{absolute_difference}}{\mu_{18}} \times 100 = \frac{0.088544174882256}{0.994794174882256} \times 100 = 8.90\%$$

Step 4 — The z -score (the -7.28 , the real alarm driver)

Percent deviation ignores *how steady Vendors normally is*. The z -score divides the gap by the trained σ , so a tiny σ makes a small gap enormous.

$$z_{18} = \frac{\text{CPI}_{18} - \mu_{18}}{\sigma_{18}} = \frac{0.90625 - 0.994794174882256}{0.01216178288304404} = \frac{-0.088544174882256}{0.01216178288304404} = -7.280525868$$

$$z_{18} \approx -7.28$$

The sign is negative because today's CPI is *below* the mean (more churn than normal). Vendors is normally extremely steady ($\sigma \approx 0.0122$), so even an 0.0885 drop lands 7.28σ away.

Step 5 — Deviation band: turn z into HIGH / MEDIUM / LOW

Rule	Level	This case
$ z > 3.0$	HIGH	$7.28 > 3.0 \Rightarrow \mathbf{HIGH} \checkmark$
$2.0 < z \leq 3.0$	MEDIUM	—
$ z \leq 2.0$	LOW	—

$|z| = 7.28$ clears the 3σ bar by more than double, so `deviation_level = "HIGH"`; and because it is the largest $|z|$ of all 18 types, it is ranked **#1**.

Step 6 — Dollar path (wCPI): why the dollar signal matches

Vendors are master-data entities with no `hasTotalAmount`, so each weight falls back to \$1. The dollar-weighted Jaccard therefore collapses onto the unweighted one:

$$\text{every } w_i = \$1 \Rightarrow \text{wCPI}_{18} = \frac{W_{\text{retained}}}{W_{\text{union}}} = \frac{58}{64} = 0.90625 = \text{CPI}_{18}$$

$$\text{dollar_}z_{18} = \frac{0.90625 - 0.994794}{0.012162} = -7.2805 \quad (\text{identical to the count } z)$$

The dollar divergence $|\text{wCPI} - \text{CPI}| = 0$, so the HIGH comes from the dollar **z -score**, not divergence: `dollar_alarm_level = "HIGH"`, `dollar_alarm = true`. This is expected for master-data types — the dollar signal duplicates the count signal rather than adding new information.

Step 7 — Composite alarm and final ranking

$$\text{composite_alarm} = (\text{deviation_level} \in \{\text{HIGH}, \text{MEDIUM}\}) \vee \text{dollar_alarm} = \text{HIGH} \vee \text{true} = \mathbf{TRUE}$$

$$\text{rank} = 1 \quad (\text{largest } |z| = 7.2805 \text{ across all 18 types})$$

What the FRP sees on the dashboard: *Vendors* (φ_{18}) — $8.9\% \cdot z \approx -7.28 \cdot \text{HIGH}$, ranked #1 of 18. Translation: vendor turnover today is 7.28 standard deviations beyond this organization's learned norm — six brand-new payees appeared at once in a category that almost never changes. That is the flagrant anomaly, surfaced for Audit-Log follow-up.

The whole chain on one line

#	Equation	Plugged in	Result
1	$n = a + b + c$	$58 + 0 + 6$	64
2	$\text{CPI} = a/n$	$58/64$	0.90625
2	$\text{AdditionRate} = c/n$	$6/64$	0.09375
3	$\%dev = \text{CPI} - \mu /\mu \times 100$	$0.088544/0.994794 \times 100$	8.90%
4	$z = (\text{CPI} - \mu)/\sigma$	$-0.088544/0.012162$	-7.281
5	$ z > 3 \Rightarrow \text{HIGH}$	$7.28 > 3$	HIGH
6	$\text{wCPI} = W_{\text{ret}}/W_{\text{union}}$	$58/64$ (\$1 each)	0.90625

#	Equation	Plugged in	Result
7	composite_alarm	HIGH $\vee$ dollar_alarm	TRUE

Sources: values read from `NGAME_Fraud_Analysis_readable_truly_clean.json` (`today_cpi_array`, `comparison_result`, `account_result`) and the `Vendors` row of `NGAME_Training_Matrix.xlsx`. Formulas verified against `ngame_churn_comparison_agent.py` (z-score, bands, dollar/composite alarms) and `ngame_cpi_analysis_agent.py` (CPI / wCPI). The a/b/c counts are reconstructed from the stated scenario (6 added, 0 removed) and the recorded CPI; the per-type counts are not stored in this JSON.